Micro I Notes

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1 Preference

1.0.1 Def "binary relation"

A binary relation R on X is a subset of $X \times X$.

The ordered pair $(x, y) \in R$ means x is (weakly) preferred to y.

1.0.2 Remark "binary relation notation"

We use $\succsim$ to denote R but all these methods are the same thing:

$$(x, y) \iff xRy \iff x \succsim y$$

1.0.3 Def "attributes of binary relations"

A binary relation $\succsim$ is

1. complete if for all $x, y \in X$ either $x \succsim y$ or $y \succsim x$
2. transitive if for all $x, y, z \in X, [x \succsim y, y \succsim z] \Rightarrow x \succsim y$
3. reflexive if for all $x \in X, x \succsim x$
4. symmetric if for all $x, y \in X, x \succsim y \iff y \succsim x$
5. irreflexive if for all $x \in X, x \not\succsim x$
6. asymmetric if for all $x, y \in X, x \succsim y \Rightarrow y \not\succsim x$
7. antisymmetric if for all $x, y \in X, [x \succsim y \text{ AND } x \neq y] \Rightarrow y \not\succsim x$
   - i.e., xRy and yRx implies $x = y$

1.0.4 "Lemma 1 (handout) rational implies"

1. $\succsim$ rational $\Rightarrow \succ$ asymmetric and transitive
2. $\succsim$ rational $\Rightarrow \sim$ symmetric and transitive
3. $\succsim$ rational $\Rightarrow x \succsim y \succsim z$ and either $x \not\sim y$ or $y \not\sim z$ implies $x \succ z$

1.0.5 Def "rational"

A binary relation $\succsim$ is *rational* if it is complete and transitive

2 Choice

2.0.1 Def "choice function"

A function $c : \mathcal{B} \to \mathcal{P}(X)$ is called a *choice function* if for all $A \in \mathcal{B}, c(A) \subseteq A$.

1. If c is a choice function, then $c(A) \neq \emptyset, \forall A \in \mathcal{B}$
2. $|c(A)| \geq 1$

2.0.2 Def "choice structure"

The pair $(\mathcal{B}, c)$ is called a *choice structure*.

2.0.3 Def "WARP"

Weak Axiom of Revealed Preference. The following are equivalent for all $A, B \in \mathcal{B}$:

- $c(A) \cap B \neq \emptyset \Rightarrow c(B) \cap A \subseteq c(A)$
- $[c(A) \cap B \neq \emptyset$ AND $c(B) \cap A \neq \emptyset] \Rightarrow [c(A) \cap B \subseteq c(B)$ AND $c(B) \cap A \subseteq c(A)]$
- $x, y \in A \cap B, x \in c(A), y \in c(B) \Rightarrow x \in c(B), y \in c(A)$
- Sen's α and β both hold

2.0.4 Def Sen's α and β

For all $A, B \in \mathcal{B}$:

1. Sen's α: If $x \in B \subseteq A$ and $x \in c(A)$ then $x \in c(B)$
2. Sen's β: If $x, y \in c(B), B \subseteq A$ and $y \in c(A)$ then $x \in c(A)$

2.0.5 Prop 1 "WARP $\iff$ α and β"

2.0.6 Term "single-valued"

We say c is single-valued if $\forall A \in \mathcal{B}, |c(A)| = 1$

2.0.7 Prop 2 "α and single-valued $\Rightarrow$ WARP"

If $(\mathcal{P}(X), c)$ satisfies α and c is single-valued, then it satisfies WARP.

3 Choice and Preference

3.0.1 Def "$c_{\succsim}$"

For any binary relation $\succsim$ define

$$c_{\succsim}(A) = \{x \in A : x \succsim y, \forall y \in A\}, \forall A \in \mathcal{P}(X)$$

3.0.2 Def "$\succsim_c$"

For a choice structure $(\mathcal{P}(X), c)$ define a binary relation $\succsim_c$ such that

$$x \succsim_c y \iff x \in c(\{x, y\})$$

3.0.3 Prop "satisfies WARP $\iff \succsim$ rational"

If $\succsim$ is a binary relation on a finite set X then $(\mathcal{P}(X), c_{\succsim})$ is a choice structure that satisfies WARP if and only if $\succsim$ is rational.

3.0.4 Prop "WARP $\Rightarrow c = c_{\succsim}$"

If $(\mathcal{P}, c)$ is a choice structure that satisfies WARP then there exists a unique rational binary relation $\succsim$ such that $c = c_{\succsim}$ (i.e., for all $A \in \mathcal{P}, c(A) = c_{\succsim}(A)$).

4 Utility

4.0.1 Def "utility function"

A function $U : X \to \mathbb{R}$ represents $\succsim$ if and only if, for all $x, y \in X$

$$x \succsim y \iff U(x) \geq U(y).$$

4.0.2 Prop "U represents $\succsim \Rightarrow$ rational"

If $U : X \to \mathbb{R}$ represents $\succsim$, then $\succsim$ is rational.

Note: this proposition does NOT require finiteness!

4.0.3 Prop "X finite + $\succsim$ rational $\Rightarrow \exists U$

If X is finite and $\succsim$ is rational then there exists $U : X \to \mathbb{R}$ that represents $\succsim$.

4.0.4 Prop "$V \iff \phi$ and ϕ strictly increasing"

Suppose $U : X \to \mathbb{R}$ represents $\succsim$. Then $V : X \to \mathbb{R}$ represents $\succsim$ if and only if there exists $\phi : U(X) \to V(X)$ such that $V = \phi \circ U$ and ϕ is strictly increasing.

5 Expected Utility

5.0.1 Def "lottery"

Suppose X is a non-empty and finite set of prizes (consequences). Then a *lottery*, p is a probability distribution over X:

$$p : X \to [0,1], \text{ such that } \sum_{x \in X} p(x) = 1.$$

We denote $\mathcal{L}$ as the set of all lotteries.

5.0.2 Term "degenerate lottery"

If $p(x) = 1$ for some $x \in X$ then we write δ_x.

5.0.3 Remark: $\mathcal{L}$ is the set of alternatives, and unlike X is infinite!

5.0.4 Def "vNM expected utility function/utility index"

$U : \mathcal{L} \to \mathbb{R}$ is an expected utility function if there exists $u : X \to \mathbb{R}$ such that $U(p) = \sum_{x \in X} u(x)p(x), \forall p \in \mathcal{L}$.

Called a "von Neumann–Morgenstern utility index" or a "Bernoulli index."

5.0.5 Def "Mixture operation"

For all $p, q \in \mathcal{L}, \alpha \in [0,1]$:

- let $\alpha p + (1-\alpha)q$ denote a lottery $r \in \mathcal{L}$ such that for all $x \in X, r(x) = \alpha p(x) + (1-\alpha)q(x)$.

5.0.6 Def "expected utility representation"

$\succsim$ on $\mathcal{L}$ has an expected utility representation if there exists $u : X \to \mathbb{R}$ such that for $p, q \in \mathcal{L}$

$$p \succsim q \iff U(p) = \sum_{x \in X} p(x)u(x) \geq U(q) = \sum_{x \in X} q(x)u(x)$$

5.0.7 Def "linear function"

A function $U : \mathcal{L} \to \mathbb{R}$ is *linear* if for all $\alpha \in [0,1]$ and $p, q \in \mathcal{L}$:

$$U(\alpha p + (1-\alpha)q) = \alpha U(p) + (1-\alpha)U(q).$$

5.0.8 Lemma "linear $\iff$ U is an expected utility function"

A function $U : \mathcal{L} \to \mathbb{R}$ is *linear* if and only if U is an expected utility function.

5.0.9 Def "vNM Axioms"

The von Neumann–Morgenstern axioms are:

- A1. (Rationality): $\succsim$ is complete and transitive.
- A2. (Independence): For all $p, q, r \in \mathcal{L}, \alpha \in (0,1)$ we have $[p \succ q] \Rightarrow [\alpha p + (1-\alpha)r \succ \alpha q + (1-\alpha)r]$.
- A3. (Continuity): For all $p, q, r \in \mathcal{L}$, if $p \succ q \succ r$ then there exists $\alpha, \beta \in (0,1)$ such that $\alpha p + (1-\alpha)r \succ q$ and $q \succ \beta p + (1-\beta)r$.
- We don't use A2* in this class, but some literature uses it:
  - A2*. (Independence, strengthened): $[p \succsim q] \iff [\alpha p + (1-\alpha)r \succsim \alpha q + (1-\alpha)r]$

5.0.10 Lemma "applying vNM Axioms"

Suppose $\succsim$ satisfies A1, A2, and A3 of vNM axioms.

(i) $\forall \alpha \in (0,1), p, q \in \mathcal{L}$:
  (a) $p \succ q \Rightarrow p \succ \alpha p + (1-\alpha)q \succ q$
  (b) $p \sim q \Rightarrow p \sim \alpha p + (1-\alpha)q \sim q$
(ii) $p \succ q, 1 \geq \alpha > \beta \geq 0 \Rightarrow \alpha p + (1-\alpha)q \succ \beta p + (1-\beta)q$
(iii) $p \sim q, \alpha \in (0,1) \Rightarrow \alpha p + (1-\alpha)r \sim \alpha q + (1-\alpha)r$
(iv) $p \succsim q, r \succsim s, \alpha \in (0,1) \Rightarrow \alpha p + (1-\alpha)r \succsim \alpha q + (1-\alpha)s$

5.0.11 Thm "A1-A3 $\iff$ EU representation"

Suppose X is finite and $\succsim$ is a binary relation on $\mathcal{L}$. Then

$$[\succsim \text{ satisfies A1, A2, and A3 }] \iff \succsim \text{ has an Expected Utility representation.}$$

5.0.12 Thm "V linear representation $\iff$ $V(p) = aU(p) + b$"

Suppose U is a linear representation of $\succsim$, and $V : \mathcal{L} \to \mathbb{R}$ is a linear function. Then

$$[V \text{ is a linear representation of } \succsim] \iff [\exists a > 0, b \in \mathbb{R} \text{ such that } V(p) = aU(p) + b, \forall p \in \mathcal{L}]$$

5.0.12.1 Term "affine" Basically a line, but the intercept doesn't have to be zero. "A function composed of a linear function and a constant and its graph is a straight line."

5.0.13 Def "simple lottery"

$p : X \to \mathbb{R}$ is a *simple lottery* if

1. $\{x \in X : p(x) > 0\}$ is finite
2. $\displaystyle\sum_{\{x \in X : p(x) > 0\}} p(x) = 1$

5.0.14 Def "simple improvement"

For all (simple) lotteries $p, q \in \mathcal{L}$, p is a simple improvement of q if there exists $\lambda \in (0,1), m, m' \in X, m > m'; , r \in \mathcal{L}$ such that

$$p = \lambda \delta_m + (1-\lambda)r \text{ and } q = \lambda \delta'_m + (1-\lambda)r.$$

5.0.15 Def "FOSD"

For $p, q \in \mathcal{L}$, p first-order stochastically dominates (FOSD) q if

1. $p \neq q$, and
2. $F_p(x) \leq F_q(x), \forall x \in X$

   caution: CDF of p SMALLER than CDF of q means p dominates q, not the other way around

5.0.16 Fact "p FOSD q $\iff$ p is a complex improvement of q"

5.0.17 Thm "EU representation means p FOSD $q \Rightarrow p \succ q$ and u strictly increasing"

Suppose $\succsim$ has an expected utility representation. Then the following are equivalent:

1. p FOSD $q \Rightarrow p \succ q$
2. p is a simple improvement of $q \Rightarrow p \succ q$
3. Every EU representation of $\succsim$ has a strictly increasing utility index ($U(p) = \sum_{x \in supp(p)} \underline{u(x)} p(x)$)

5.0.18 Def "Betweenness Axiom"

$$\forall p, q \in \mathcal{L}, \lambda \in (0,1):$$
$$p \sim q \Rightarrow \lambda p + (1-\lambda)q \sim p$$

5.0.19 Def "Allais paradox"

Individuals prefer certainty over a risky outcome even if this defies the expected utility axiom

5.1 Risk Attitudes

5.1.1 Lemma

Suppose $\succsim$ has an expected utility representation with a strictly increasing and continuous index u. Then $\forall p \in \mathcal{L}, \exists$ unique $CE(p) \in X$ such that $p \sim \delta_{CE(p)}$. In this case, $CE(p)$ is the *certainty equivalent* of p.

$$CE(p) = u^{-1}(U(p)) = u^{-1}(\sum_{x \in supp(p)} u(x)p(x)) \in X \iff u(CE(p)) = U(p)$$

5.1.2 Def "risk attitudes"

Let $\bar{p} = \sum_{x \in supp(p)} p(x)x$ be the expected value of p. Then p is riskier than $\delta_{\bar{p}}$. For all $p \in \mathcal{L}$, $\succsim$ is

1. *risk averse* if $\delta_{\bar{p}} \succsim p$
2. *risk neutral* if $\delta_{\bar{p}} \sim p$
3. *risk seeking* if $p \succsim \delta_{\bar{p}}$

5.1.3 Def "simple Mean-Preserving Spread (MPS)"

Say $\hat{p} = \lambda p + (1-\lambda)r$ is a simple mean-preserving spread of $\hat{q} = \lambda\delta_{\bar{p}} + (1-\lambda)r$. $\hat{p}$ is riskier than $\hat{q}$.

5.1.4 Thm "risk-averse $\iff$ u concave $\iff$ MPS"

$\succsim$ has an EU representation such that its utility index, u, is strictly increasing and continuous. Then the following are equivalent:

1. $\succsim$ is risk-averse / -seeking / -neutral
2. $[p$ is a simple MPS of $q] \Rightarrow q \succsim p$
3. u is concave / convex / affine

5.1.5 Def "more risk averse"

$\succsim_1$ is more risk averse than $\succsim_2$ if for all $m \in X, p \in \mathcal{L}$ we have

$$\delta_m \succsim_2 p \Rightarrow \delta_m \succsim_1 p.$$

5.1.5.1 Term "Absolute risk aversion"

Call $-\frac{u_1''}{u_2'}$ Arrow-Pratt's measure of absolute risk aversion (ARA).

5.1.6 Thm "more risk averse $\iff$ concave composition"

$\succsim_1, \succsim_2$ have an EU representation with u_1, u_2 strictly increasing and smooth. Then the following are equivalent:

1. $\succsim_1$ is (always) more risk averse than $\succsim_2$.
2. there exists a concave and strictly increasing function $f : u_2(X) \to u_1(X)$ such that $u_1 = f \circ u_2 = f(u_2(x)), \forall x \in X$
3. $ARA_{u_1} \geq ARA_{u_2}$

5.1.6.1 Term "smooth" Twice continuously differentiable

5.1.7 Def "CARA"

$$ARA = -\frac{u''(x)}{u'(x)} = a(\text{const} \in \mathbb{R})$$

Constant Absolute Risk Aversion (CARA) means risk attitude does not depend on the initial wealth level.

5.1.8 Def "functional form for CARA"

$$u(x) = \begin{cases} \frac{1-e^{-ax}}{a} \text{ if } a \neq 0 \\ x \text{ if } a = 0 \end{cases}$$

- $a > 0 \Rightarrow$ risk-averse
- $a = 0 \Rightarrow$ risk-neutral
- $a < 0 \Rightarrow$ risk-seeking

  Can compare the "a-level" of individuals to see who is more risk averse (e.g., $a_1 > a_2 \Rightarrow$ person 1 more risk-averse)

5.1.9 Def "CRRA"

$$RRA = -x\frac{u''(x)}{u'(x)} = \text{const} = r$$

Constant Relative Risk Aversion (CRRA) means risk attitude does not depend on inflation.

$$u(x) = \begin{cases} \frac{x^{1-r}}{1-r} \text{ if } r \neq 1 \\ \ln x \text{ if } r = 1 \end{cases}$$

person is risk-neutral when $u(x) = x$.

6 Demand Theory

6.0.1 Def "Well-behaved" preference

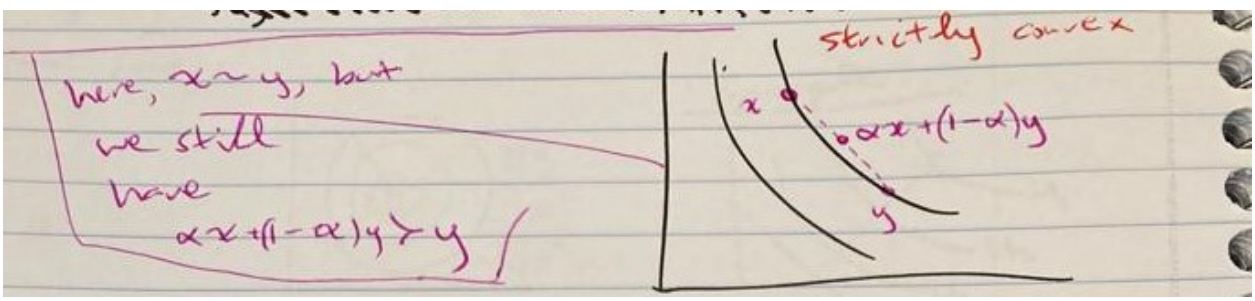
1. Monotone
$$[x_i \geq y_i, \forall i] \Rightarrow [x \succsim y]$$
2. Strongly Monotone
$$[x_i \geq y_i, \forall i \text{ and } x \neq y] \Rightarrow [x \succ y]$$
3. MWG Monotone
$$[x_i > y_i, \forall i] \Rightarrow [x \succ y]$$
4. Locally non-satiated
$$\forall x \in X, \forall \varepsilon > 0, \exists y \in B_\varepsilon(x) \text{ such that } y \succ x$$
5. Continuity $x^{(m)} \to x$ and $y^{(m)} \to y, \forall m \geq 1$ and $x^{(m)} \succsim y^{(m)} \Rightarrow x \succsim y$

   $2 \Rightarrow 3 \Rightarrow 4$ (but 3 alone does not imply 4)

6.0.2 Def "convex relation"

For all $x, y \in X$ and for all $\alpha \in (0, 1)$:

- $\succsim$ is *convex* if
$$x \succsim y \Rightarrow \alpha x + (1-\alpha)y \succsim y$$
- $\succsim$ is *strictly convex* if
$$x \succsim y, x \neq y \Rightarrow \alpha x + (1-\alpha)y \succ y$$



6.0.3 Claim "$\succsim$ is rational $\not\Rightarrow \succsim$ has a utility representation"

6.0.4 Term "lexicographic preferences"

1. $\succsim$ is rational
2. $\succsim$ is not continuous
3. $\succsim$ does not have a utility representation

6.0.5 Def "economic preference / well-behaved preference"

A binary relation $\succsim$ is an *economic preference* (well-behaved preference) if it is 1. rational, 2. continuous, 3. monotone, and 4. locally non-satiated.

6.0.6 Thm "economic preference $\Rightarrow$ represented by continuous U"

$\succsim$ is an economic preference $\Rightarrow \succsim$ can be represented by a continuous utility function U.

6.0.7 Remark U is also monotone and MWG monotone, but not strong

7 Demand

7.0.0.1 Term "price vector" $P \in \mathbb{R}^n_{++}$. $p \in P : p = (p_1, ..., p_n), p_i > 0 \forall i$

7.0.0.2 Term "Budget constraint"

$$B(p, m) = \{\vec{x} \in X : p \cdot \vec{x} = \sum_{i=1}^{n} p_i \cdot x_i \leq m\}, \forall p \in P, m \geq 0.$$

$B(p, m)$ is compact, non-empty (and convex, but that isn't important for now.)

7.0.1 Thm "economic preference $\Rightarrow$ solution exists"

If $\succsim$ is an economic preference then $\forall p, m \geq 0, \exists x^* \in B(p, m)$ such that

1. $x^* \succsim y, \forall y \in B(p, m)$ and
2. $p \cdot x^* = m$.

7.0.2 Def "demand"

$$D : \mathcal{B} \to \mathcal{P}(X) \text{ such that } D(B(p, m)) = \{x^* \in B(p, m) : x^* \succsim y \forall y \in B(p, m)\}$$

Where $\mathcal{B} = \{B(p, m) : p \in P, m \geq 0\} \subsetneq \mathcal{P}(X)$. We can verify that $(\mathcal{B}, D)$ is a choice structure.

7.0.3 Def "demand function"

If $D(B(p, m))$ is always a singleton we can define:

$$x : P \times \mathbb{R}_+ \to X$$

such that $x(p, m)$ is the only element of $D(B(p, m))$. We call $x(\cdot)$ the demand function.

7.0.4 Thm "strictly convex $\Rightarrow$ demand fn well-defined"

If $\succsim$ is a strictly convex economic preference then $x(\cdot)$ is a well-defined and continuous function.

7.0.5 Def "indirect utility function"

$V : P \times \mathbb{R}_+ \to \mathbb{R}$ such that $V(p, m) = U(x(p, m))$.

7.0.6 Def "neoclassical utility function"

$U : X \to \mathbb{R}$ is a *neoclassical utility function* if U

1. is continuous, and
2. represents a strictly convex economic preference.

7.0.7 Thm "neoclassical $U \Rightarrow$ properties of V"

Suppose U is neoclassical and V is the associated indirect utility function. Then,

1. V is continuous,
2. V is strictly increasing in m,
3. V is quasiconvex ($V(p, m) \geq V(p', m') \Rightarrow V(p, m) \geq V(\alpha p + (1 - \alpha)p', \alpha m + (1 - \alpha)m')$, and
4. V is homogeneous of degree 0 ($V(p, m) = V(\lambda p, \lambda m)$)

7.1 Hicksian Demand

7.1.1 Thm "neoclassical $\Rightarrow$ unique minimization solution"

Suppose U is neoclassical. Then for all $p \in P, u \in U(X), \exists$ unique $x^* \in X$ such that

1. $U(x^*) = u$ and
2. $\forall x \in X, x \neq x^* : U(x) \geq u \Rightarrow p \cdot x > p \cdot x^*$

7.1.2 Def "Hicksian demand function

Let $x^h(p, u)$ denote $x^* \forall p \in P, u \in U(X)$. Then $x^h : P \times U(X) \to X$ is called the *Hicksian demand function.*

7.1.3 Def "expenditure function"

Define the *expenditure function*, $e : P \times U(X) \to X$ such that $\forall p \in P, u \in U(X)$:

$$e(p, u) = p \cdot x^h(p, u).$$

7.1.4 Thm "duality theorem"

Suppose U is neoclassical. Then

1. $e(p, V(p, m)) = m$
2. $V(p, e(p, u)) = u$
3. $x(p, e(p, u)) = x^h(p, u)$
4. $x^h(p, V(p, m)) = x(p, m)$.

Three important results:

7.1.5 Thm "relating $e(\cdot)$ and $x^h(\cdot)$"

Suppose U is neoclassical. Then

1. $e(\cdot)$ is continuous,
2. $e(\cdot)$ is strictly increasing in u and non-decreasing in p_i,
3. $e(\cdot)$ is concave in p for fixed $u \in U(X)$,
4. $e(\lambda p, u) = \lambda e(p, u), \forall \lambda > 0$, and
5. $\frac{\partial}{\partial p_i} e(p, u) = x_i^h(p, u), \forall i = 1, ..., n$.
   - This is Shephard's Lemma. It's the counterpart to Roy's identity, just on the Hickisan side (and simpler).

7.1.6 Thm "relating $x(\cdot)$ and $V(\cdot)$ (Roy's identity)"

Suppose U is neoclassical, $V(p, m)$ is differentiable at (p, m) and $x(p, m)$, is in the interior of X. Then

$$x_i(p, m) = -\frac{\frac{\partial}{\partial p_i} V(p, m)}{\frac{\partial}{\partial m} V(p, m)}$$

7.1.7 Thm "relating $x(\cdot)$ and $x^h(\cdot)$ (Slutsky equation)"

Suppose U is neoclassical, $x(\cdot)$ is continuously differentiable, and $x(p, m)$ is in the interior of X. Then

$$\frac{\partial x_i^h}{\partial p_k}(p, V(p, m)) = \underbrace{\frac{\partial x_i}{\partial p_k}(p, m)}_{\text{substitution effect}} + \underbrace{\frac{\partial x_i}{\partial m}(p, m) x_k(p, m)}_{\text{income effect}}, \forall i, k \in \{1, ..., n\}$$

i, k can be same (sensitivity of demand to price of own good) or different (sensitivity of demand to price of another good)

8 Misc

8.0.1 "Utility Maximization"

Utility is maximized when

$$\frac{MU_x}{p_x} = \frac{MU_y}{p_y}$$

Note that the objective function is linear in t. Thus, maximization occurs at a "corner," where either $t = 0$ or $t = 1$.If $a_1 > a_2$, then it is best to set $t = 1$; otherwise, it is best to set $t = 0$.

8.1 Algorithms

8.1.1 "Hicksian..."

finish

8.1.2 "Random tips/facts"

TBD

8.1.3 "Mistakes to avoid"

TBD